Media Redundancy Protocol



A Leading Provider of Smart, Connected and Secure Embedded Control Solutions



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MRP - Introduction

What is it: Media Redundancy Protocol(MRP) is a recovery protocol based on a ring topology

Who did it: MRP is standardized by the International Electrotechnical Commission as IEC 62439-2

Why we do it: Needed in industrial networks

Why kernel space: Changes are needed to implement this

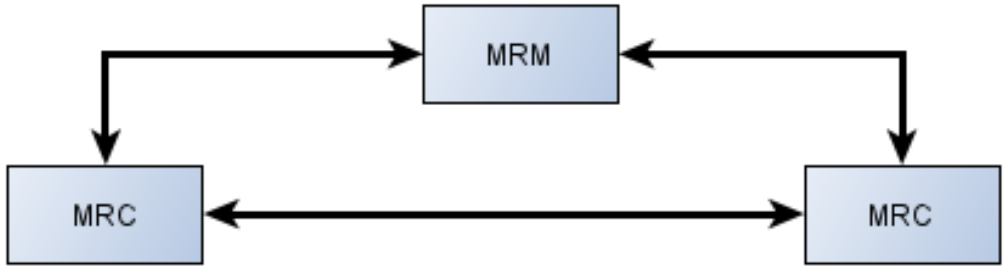
Why so complicated: You will see soon

Status: MRM/MRC/MRA accepted, continue with MIM/MIC.

What we want: Help review, and guide on doing the best possible integration

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Roles

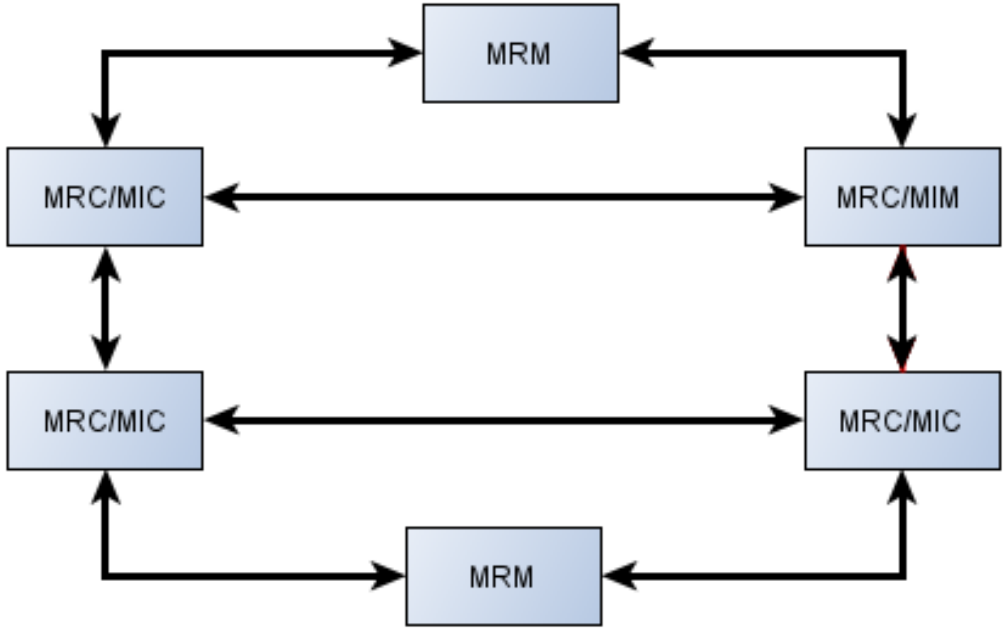


Single Ring:

MRM: Manager

MRC: Client

MRA: Automanager



Connecting Rings

MIM: Interconnection Manager

MIC: Interconnection Client

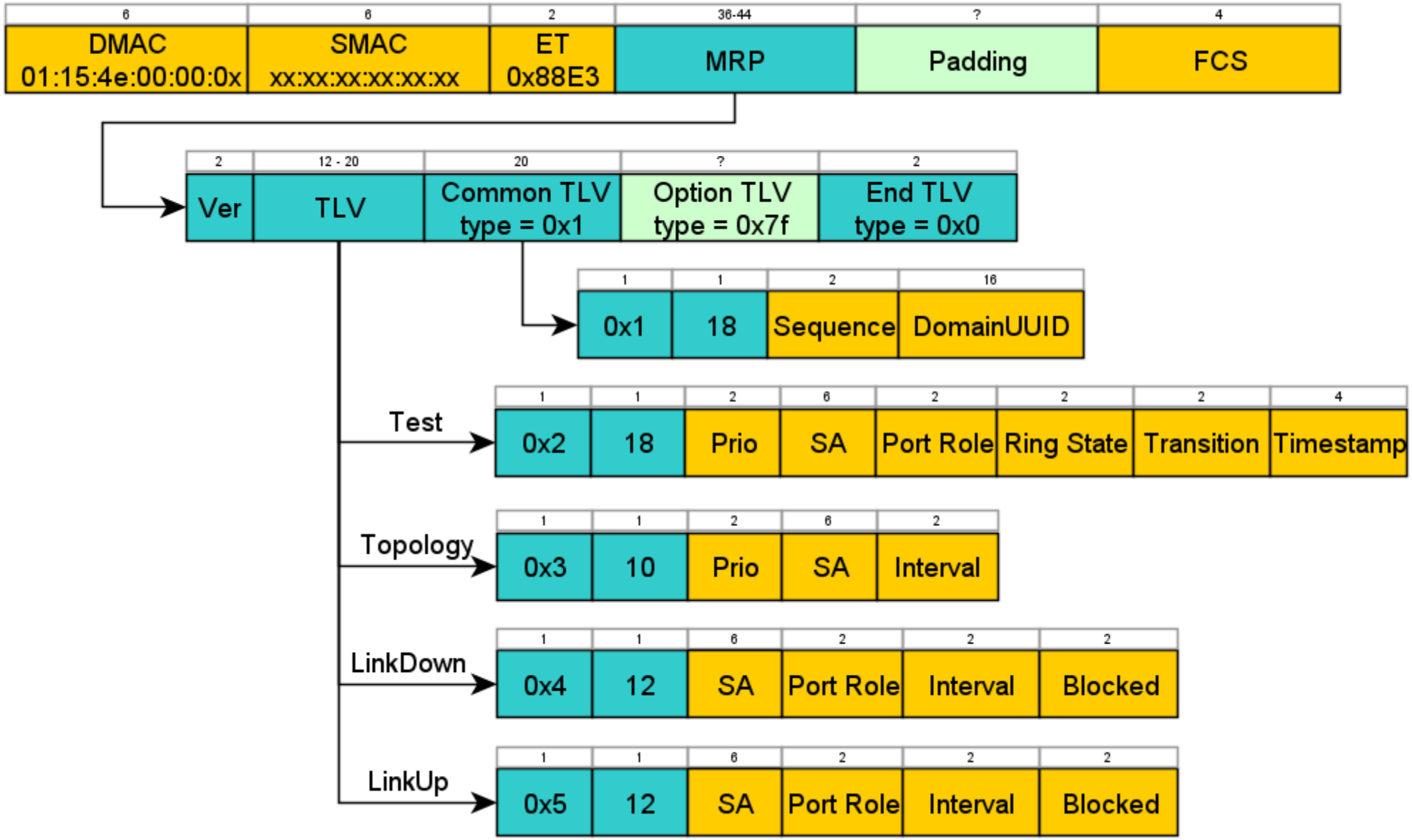
Interconnect has two modes:

LC: CFM Based

RC: MRP_InTest frames

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PDU types



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Who does what?

		PDU-Types			
		Test	**Topology**	**LinkDown/Up**	**Option**
ROLE	**MRM**	Generate Terminate	Generate Terminate	Trap Terminate	Terminate
	MRC	Forward	Forward Trap	Generate Forward	Forward
	MRA	Listen	Forward	Forward	Generate Forward

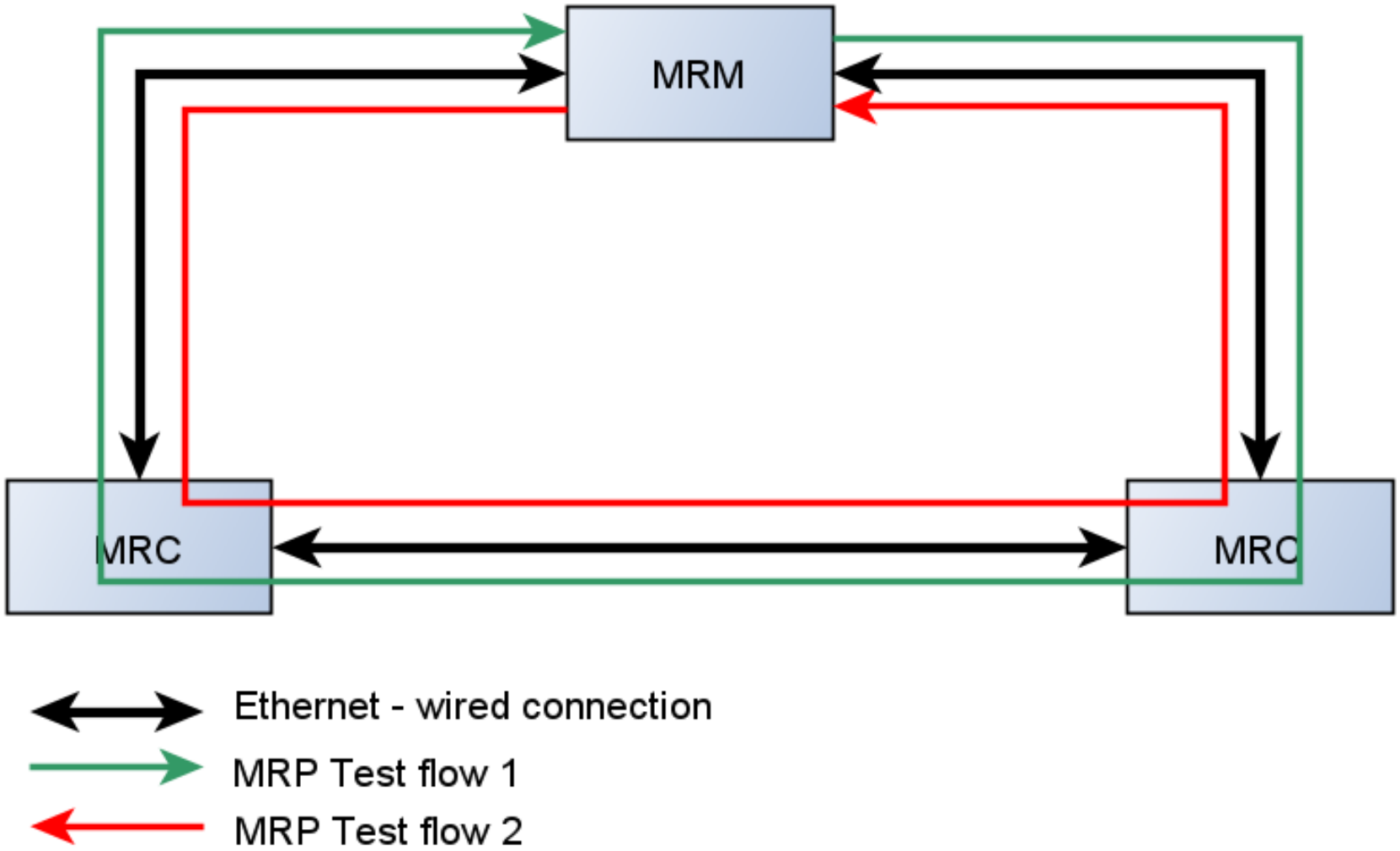
Offload able in MRP aware HW

Offload-able in Non-MRP aware HW – but require some TCAM

We see no need for HW offload - SW only

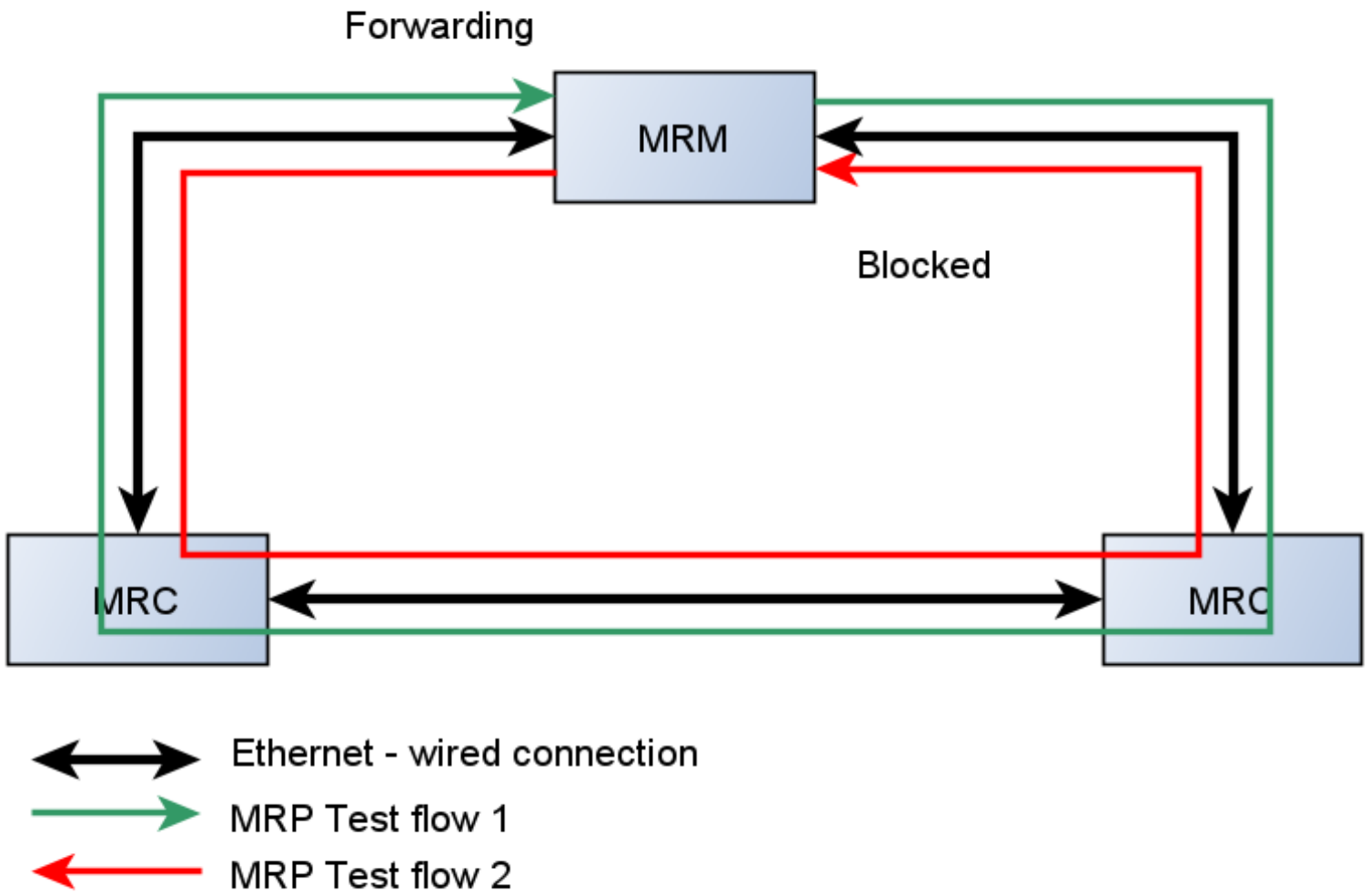
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MRM/MRC

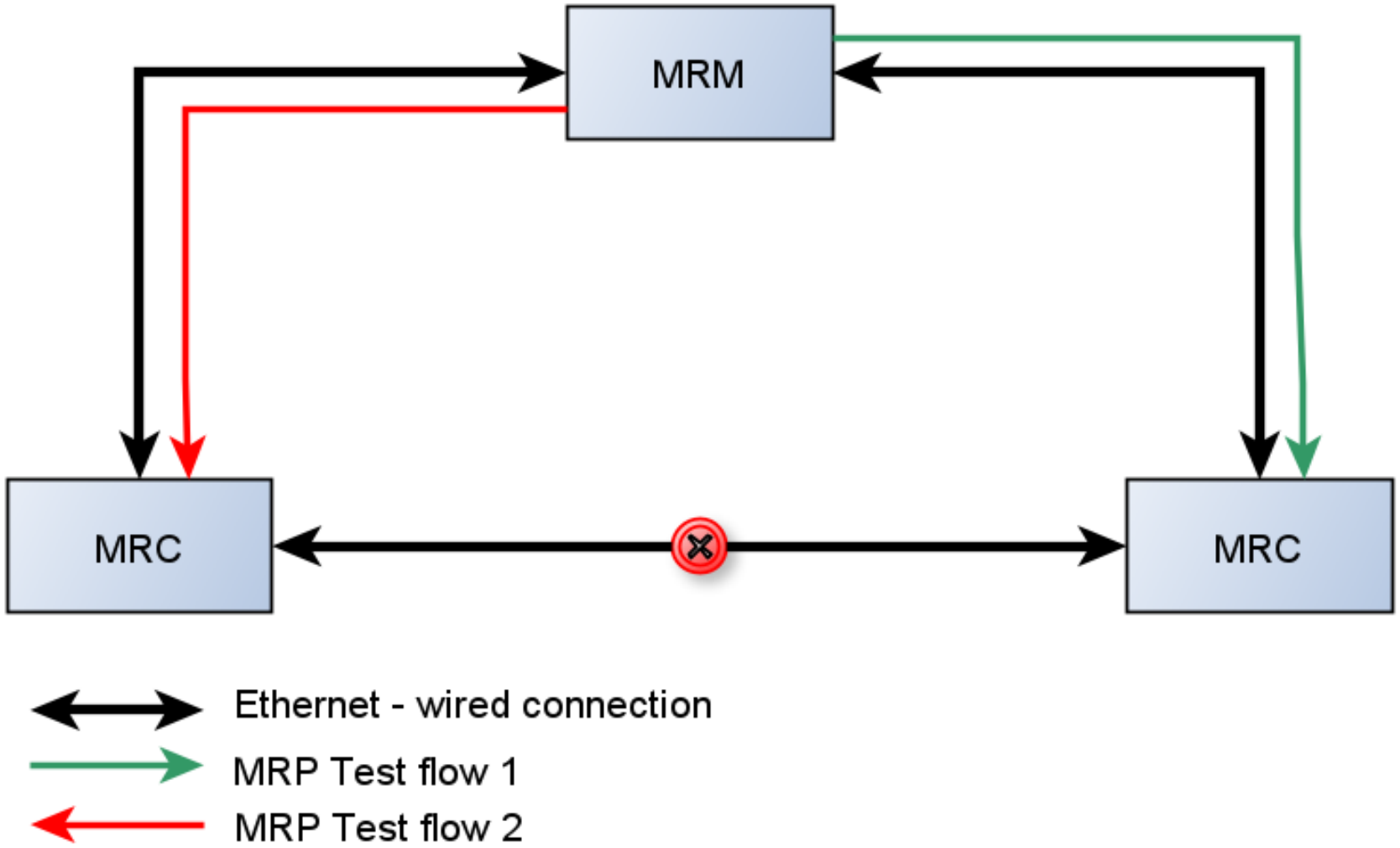


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MRM/MRC

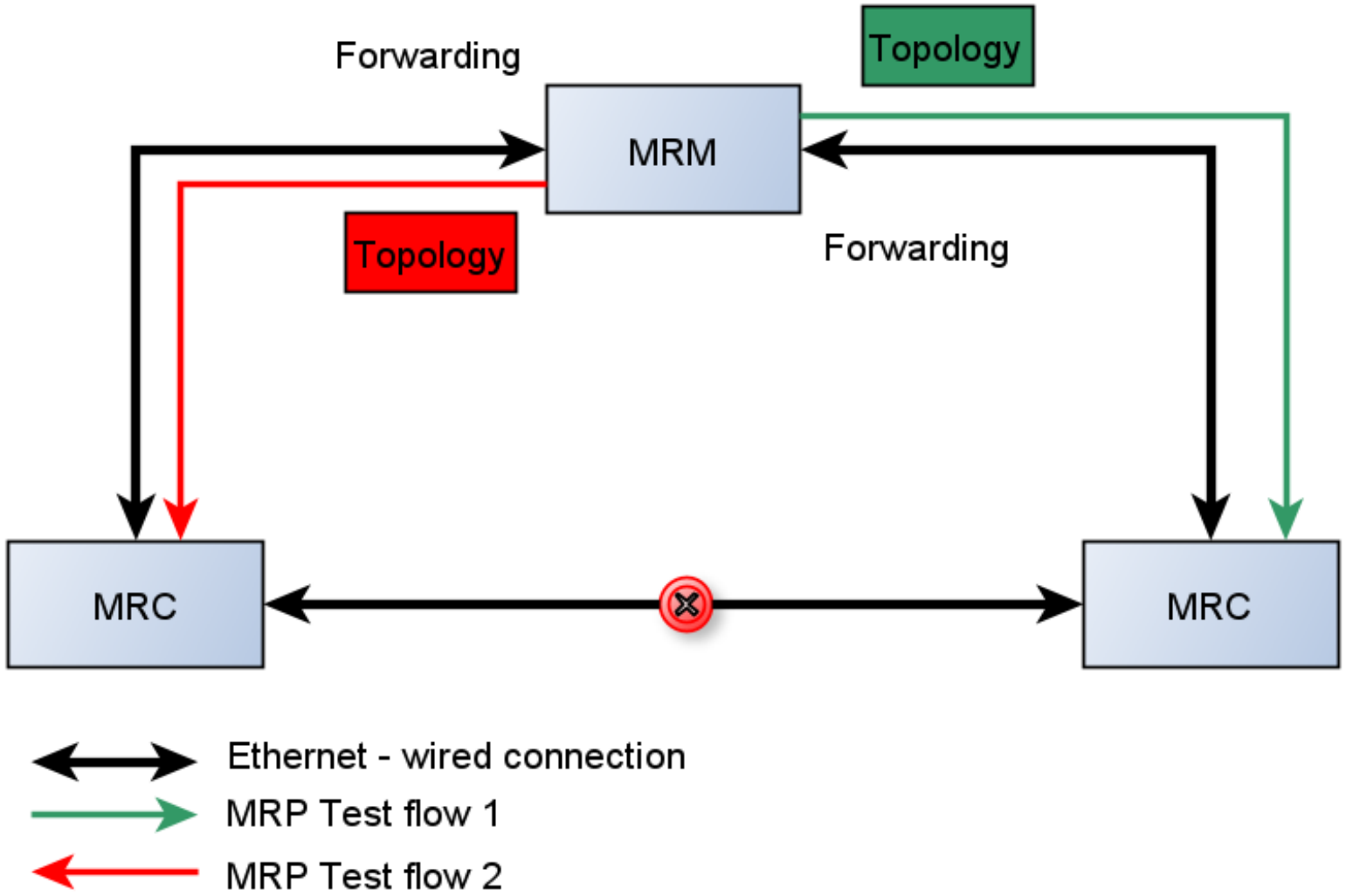

Forwarding
MRM
Blocked
MRC
MRC
Ethernet - wired connection
MRP Test flow 1
MRP Test flow 2

MRM/MRC



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MRM/MRC

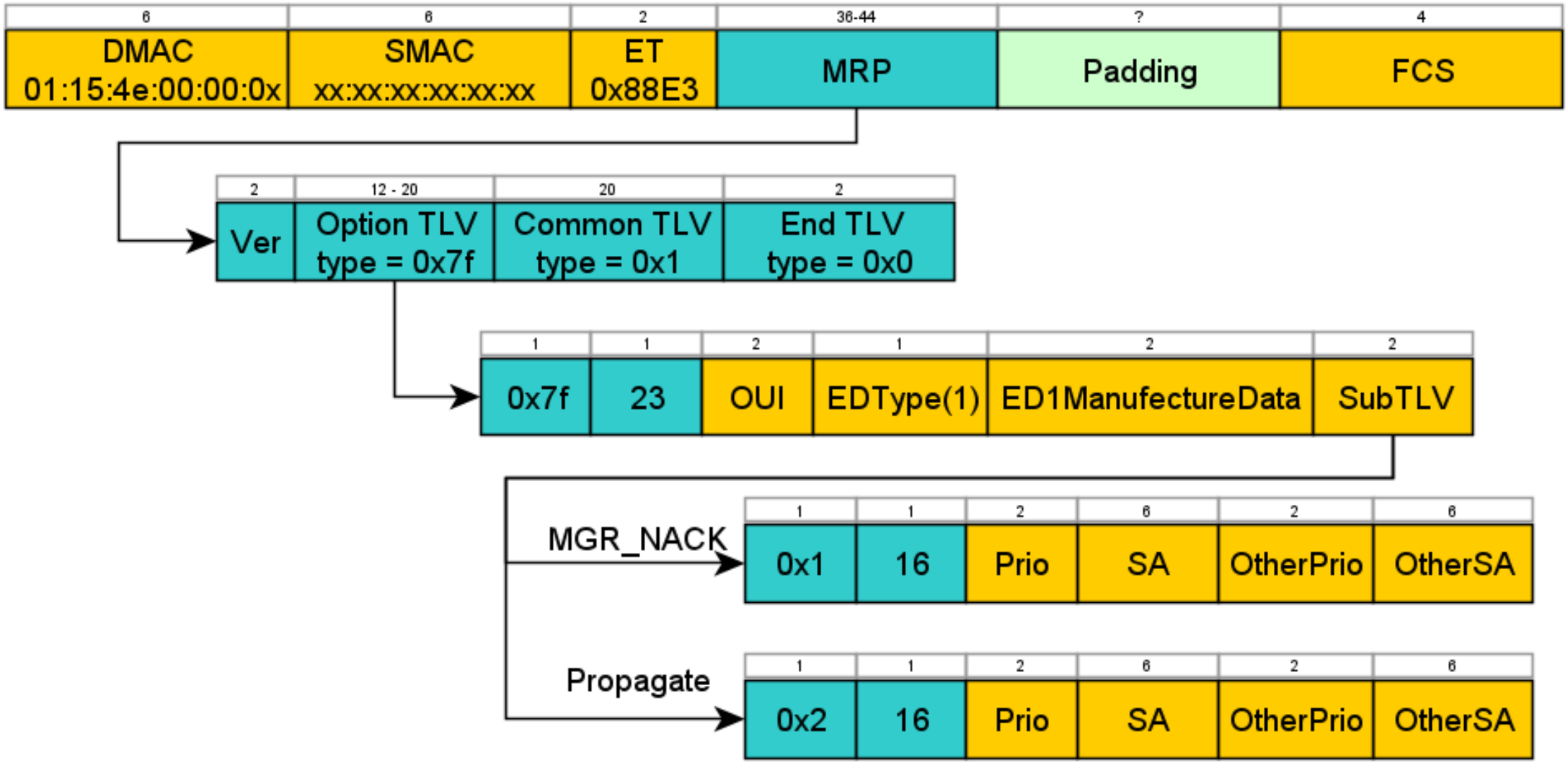


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Auto Manager(MRA)

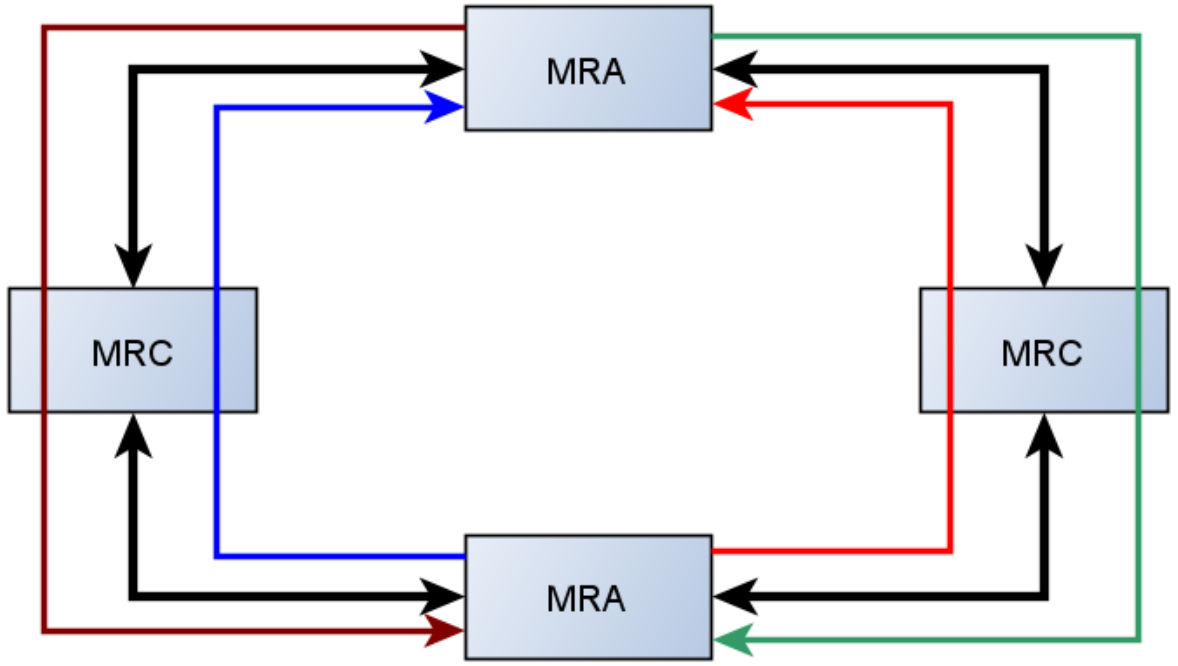
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Auto Manager PDU



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MRA



Ethernet - wired connection

MRP Test flow 1

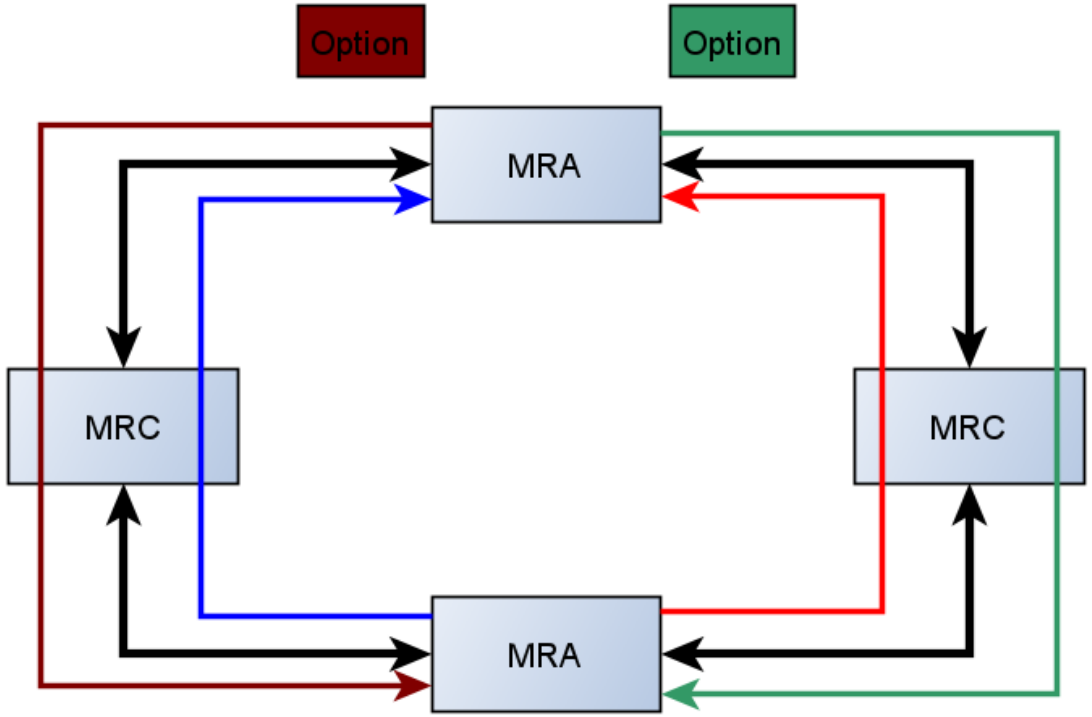
MRP Test flow 2

MRP Test flow 3

MRP Test flow 4

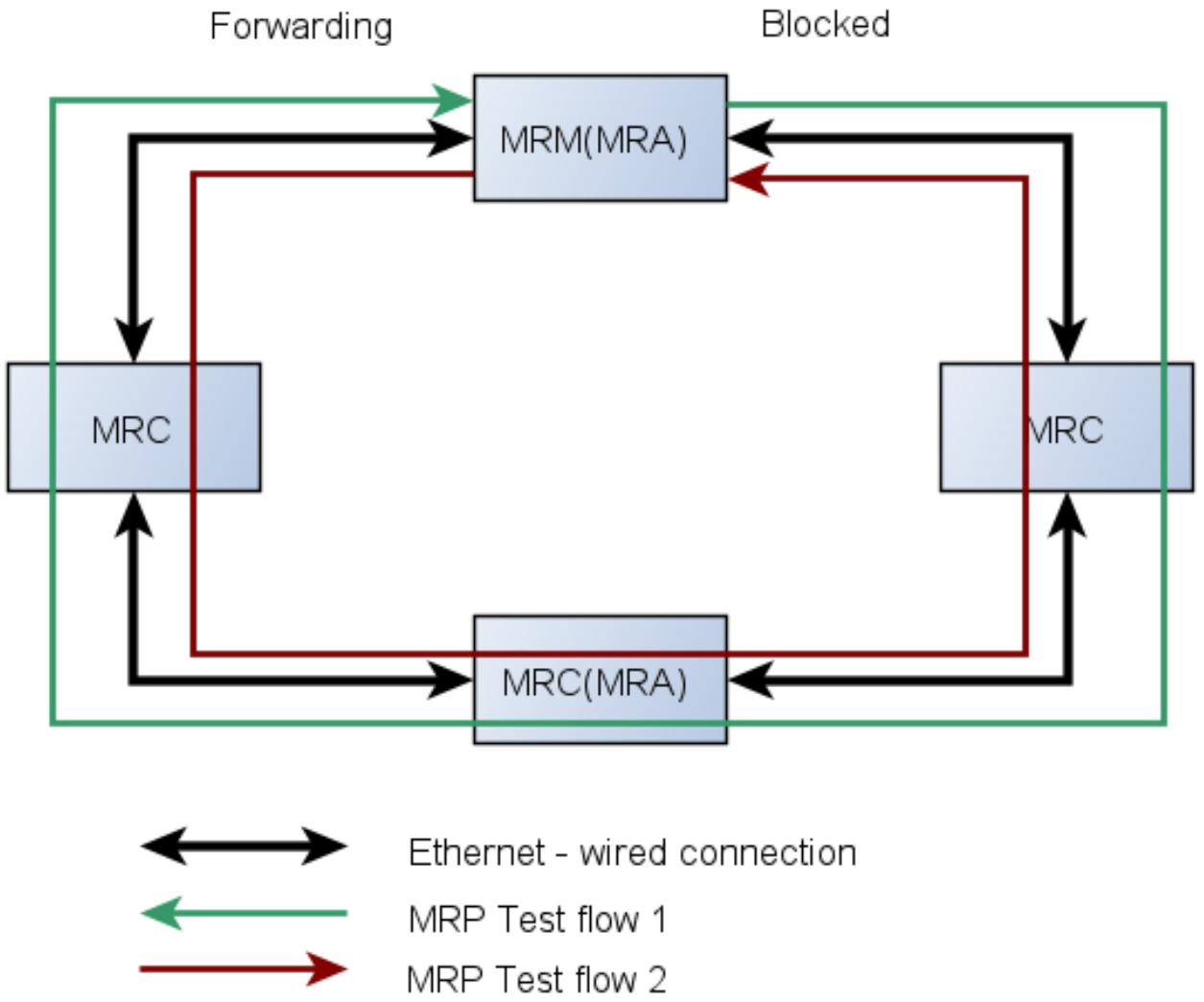
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MRA



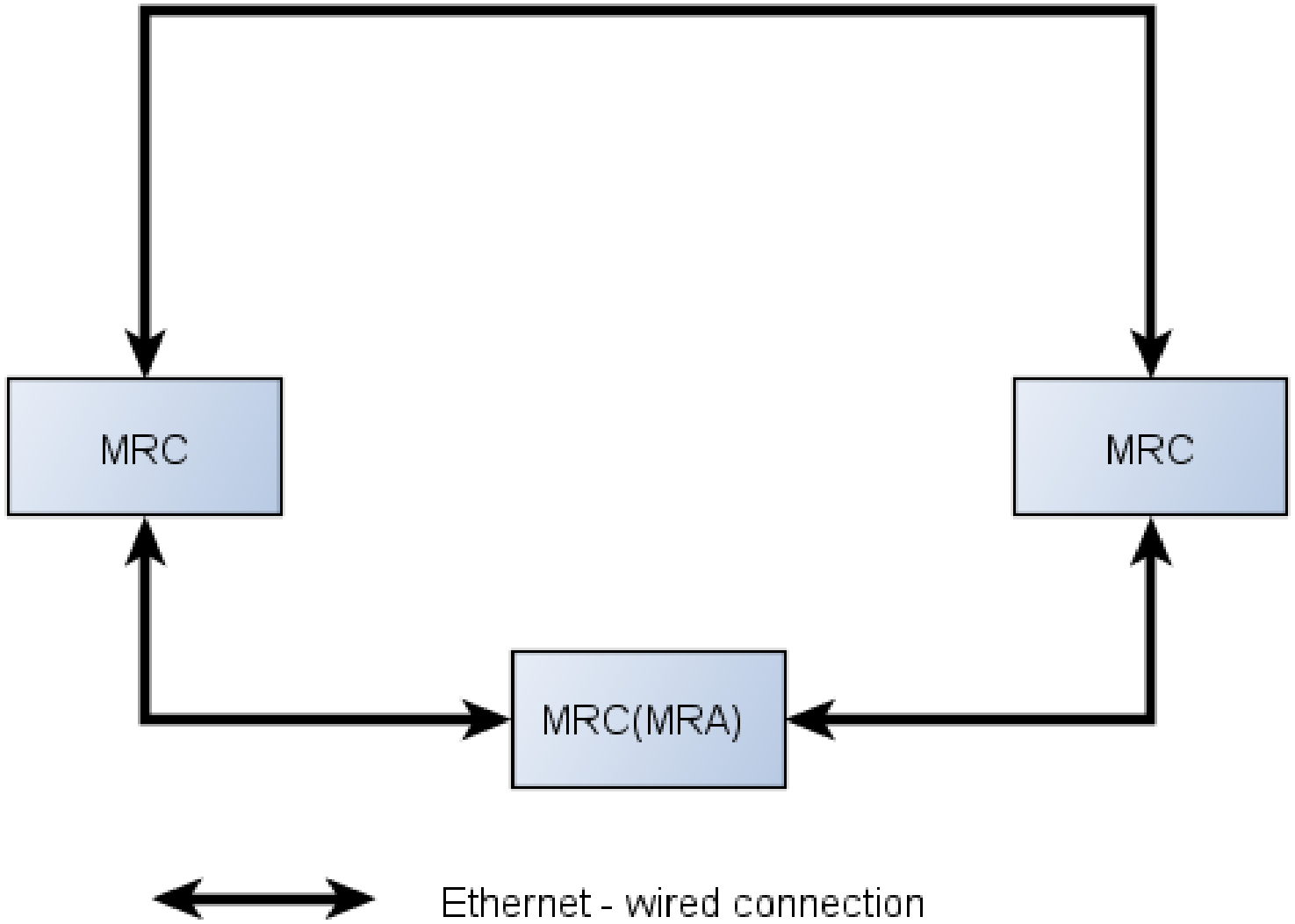
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MRA

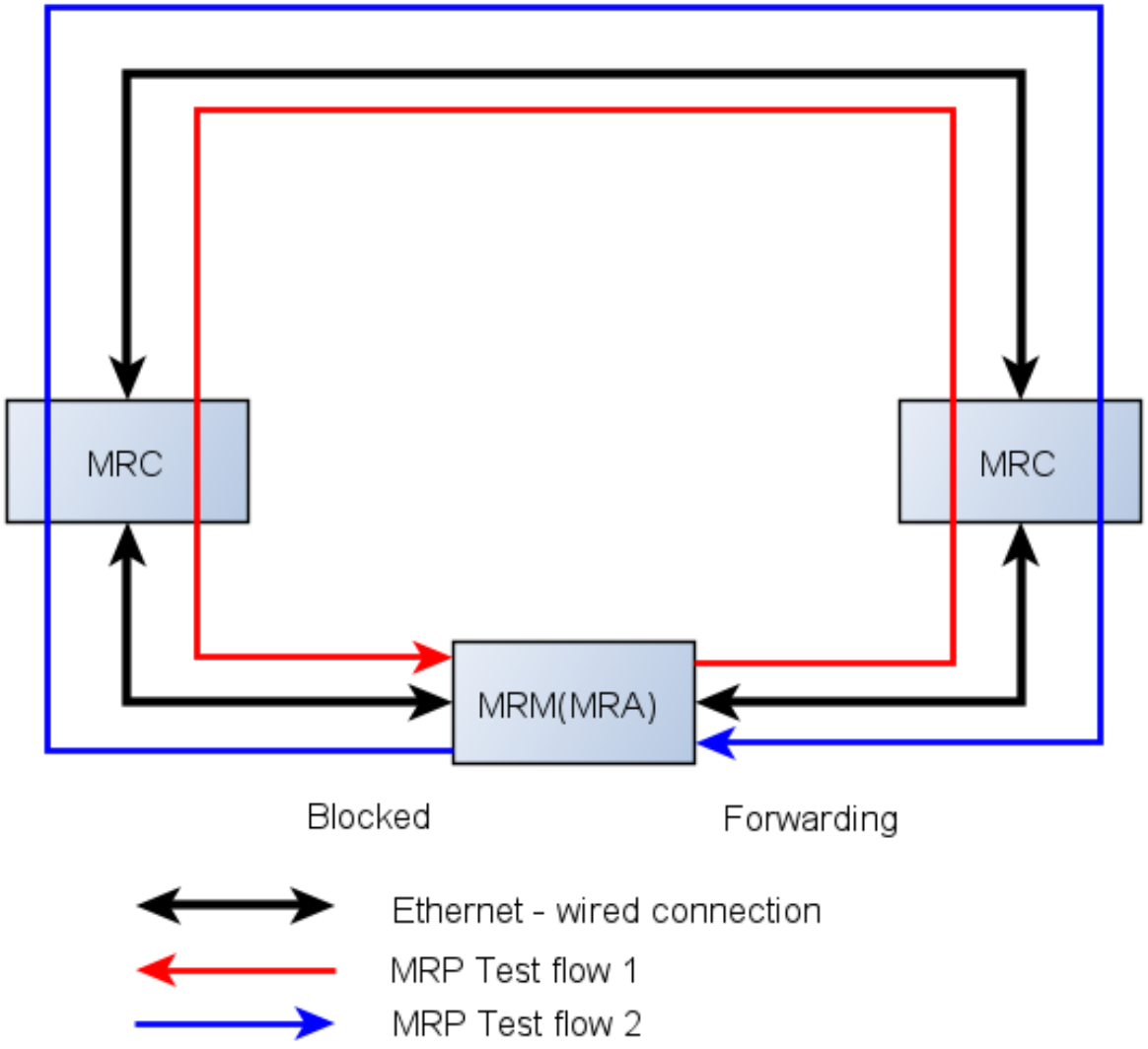


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MRA



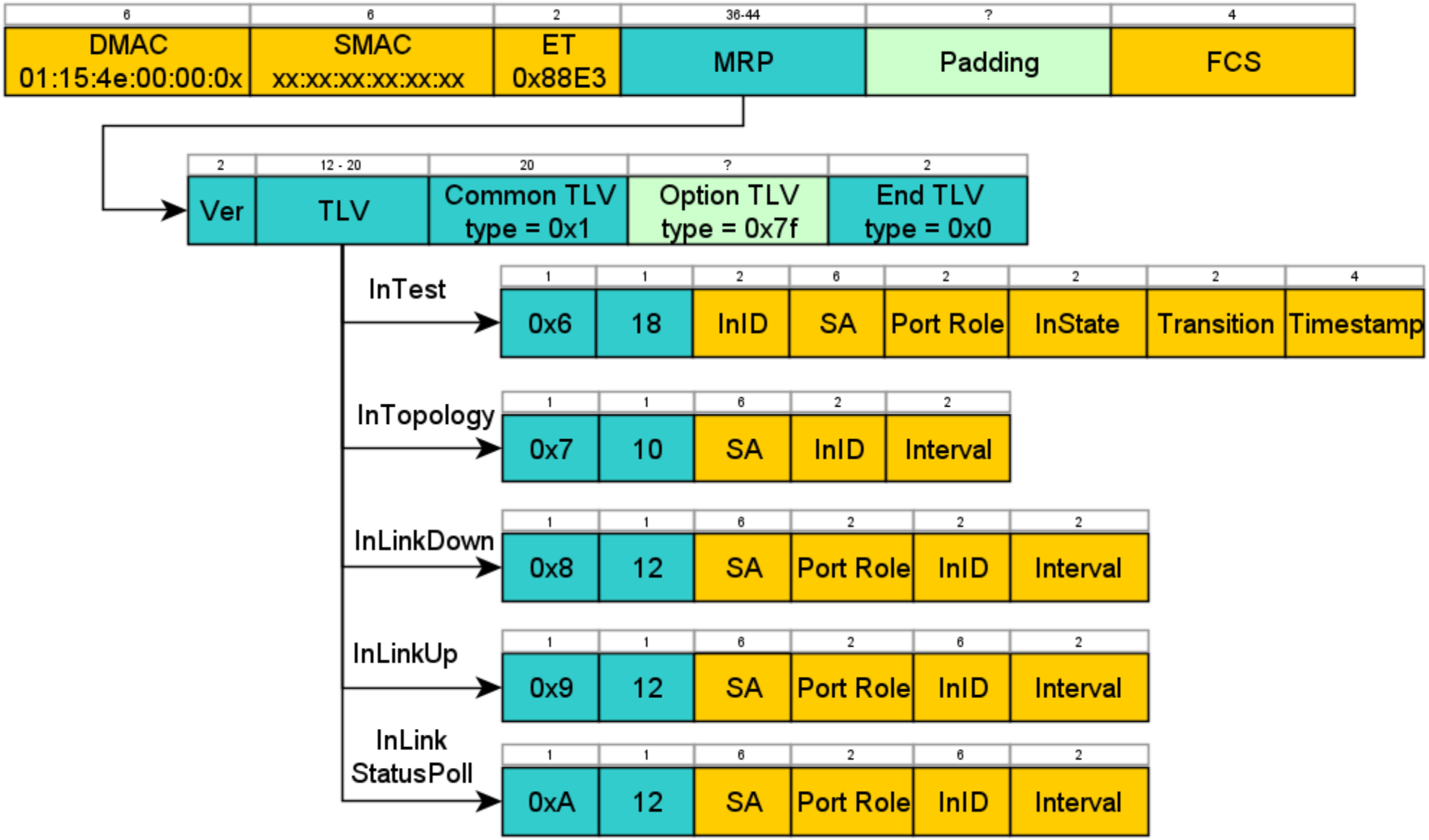
MRA


MRC
MRC
MRM(MRA)
Blocked
Forwarding
Ethernet - wired connection
MRP Test flow 1
MRP Test flow 2

Interconnect (MIC/MIM)
Connecting rings with rings

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Interconnect PDU



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Who does what?

		PDU Types			
		InTest	**InTopology**	**InLinkUp/ Down**	**InLink-StatusPoll**
ROLE	**MIM**	Generate Terminate	Generate	Trap	Generate
			Terminate	Terminate	Terminate
	MIC	Forward	Forward	Generate	Forward
				Forward	
	MRM	Terminate or Forward	Trap or Forward	Terminate or Forward	Terminate or Forward
	MRC	Forward	Forward	Forward	Forward

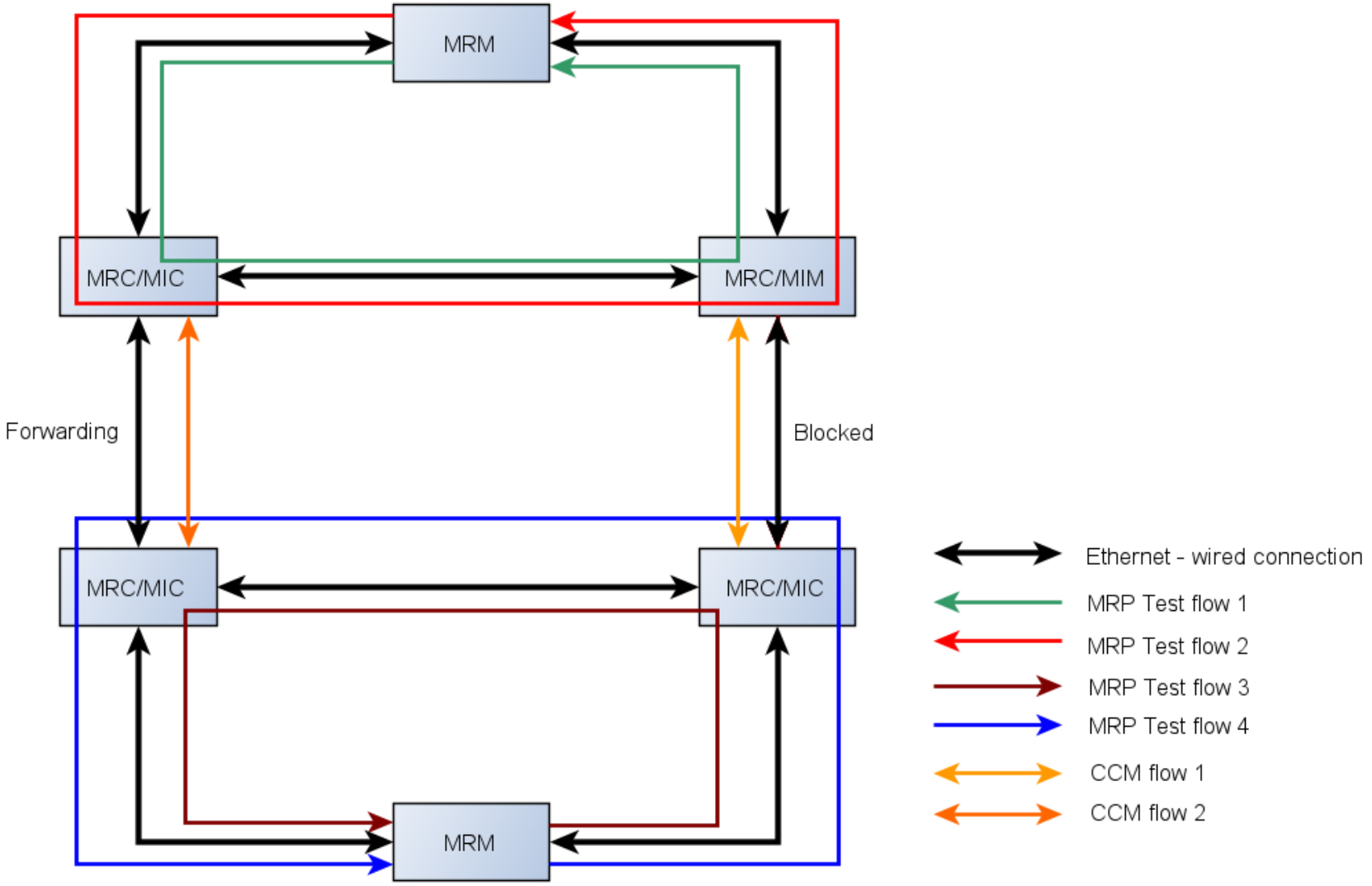
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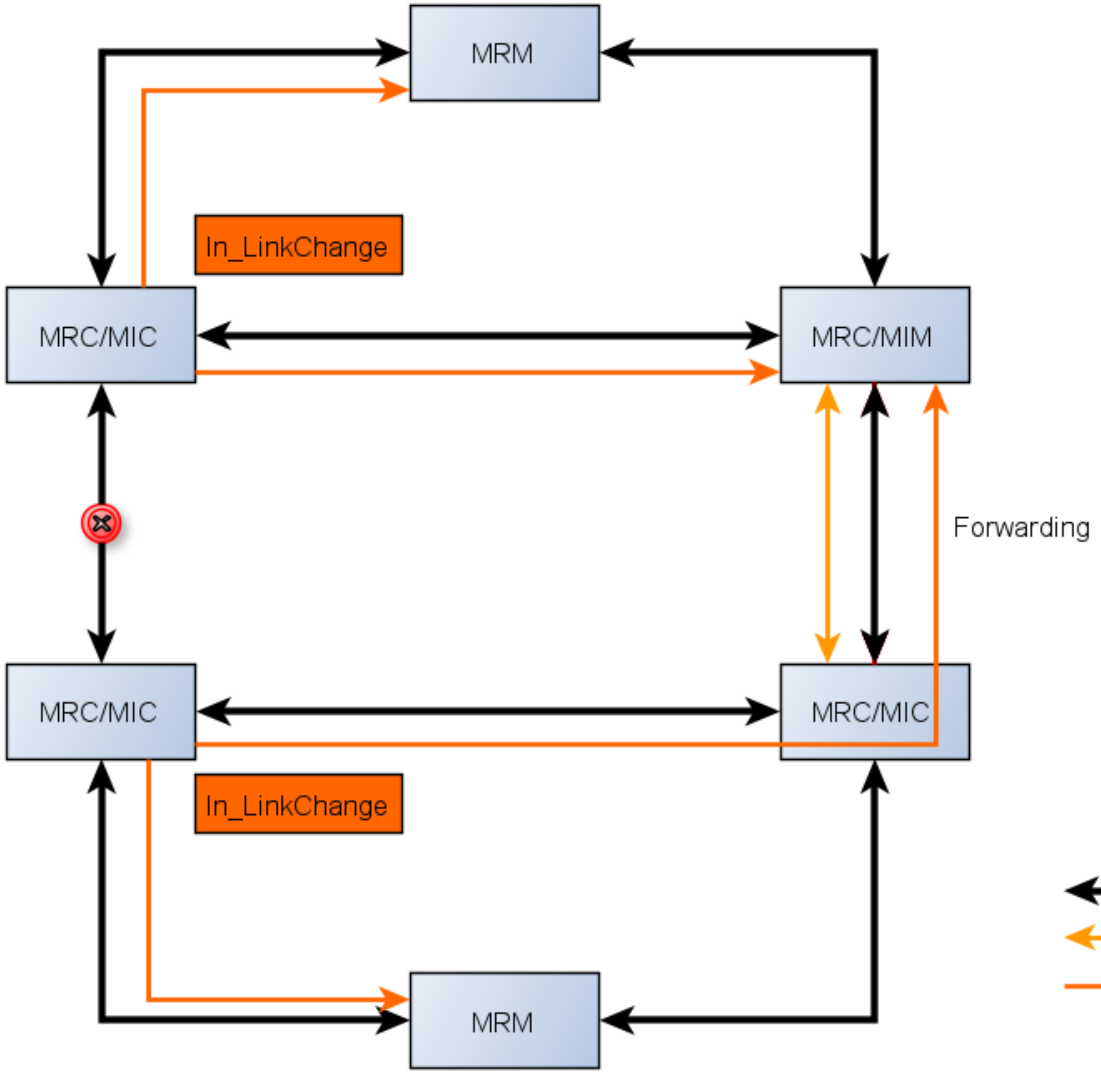
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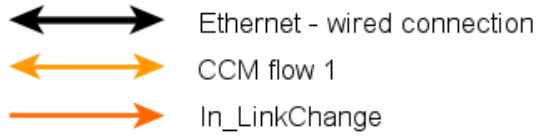
MIM/MIC - LC


MRM
MRC/MIC
MRC/MIM
Forwarding
Blocked
MRC/MIC
MRC/MIC
MRM
Ethernet - wired connection
MRP Test flow 1
MRP Test flow 2
MRP Test flow 3
MRP Test flow 4
CCM flow 1
CCM flow 2

MIM/MIC - LC

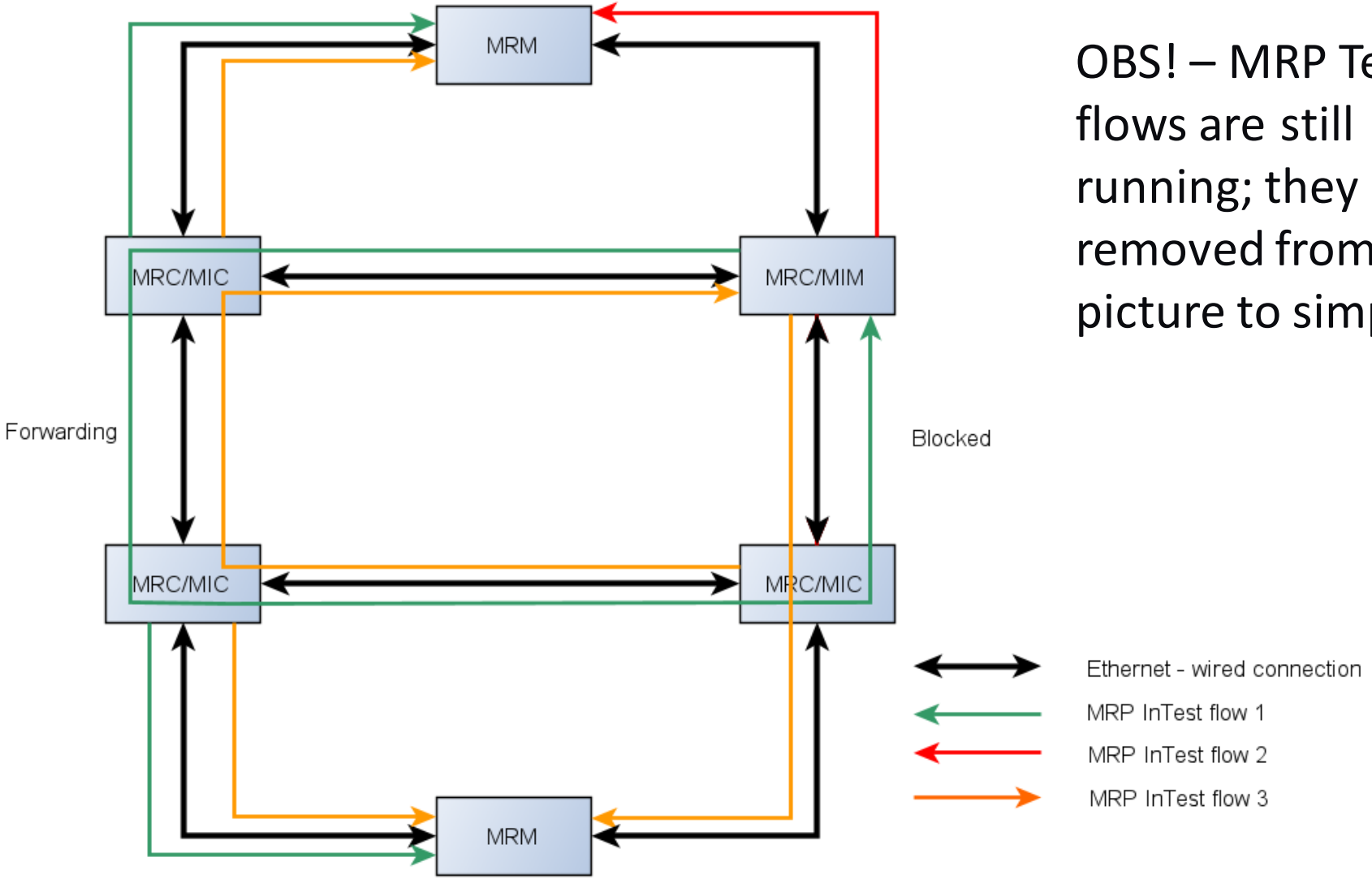


OBS! – MRP Test flows are still running; they are just removed from picture to simplify it.



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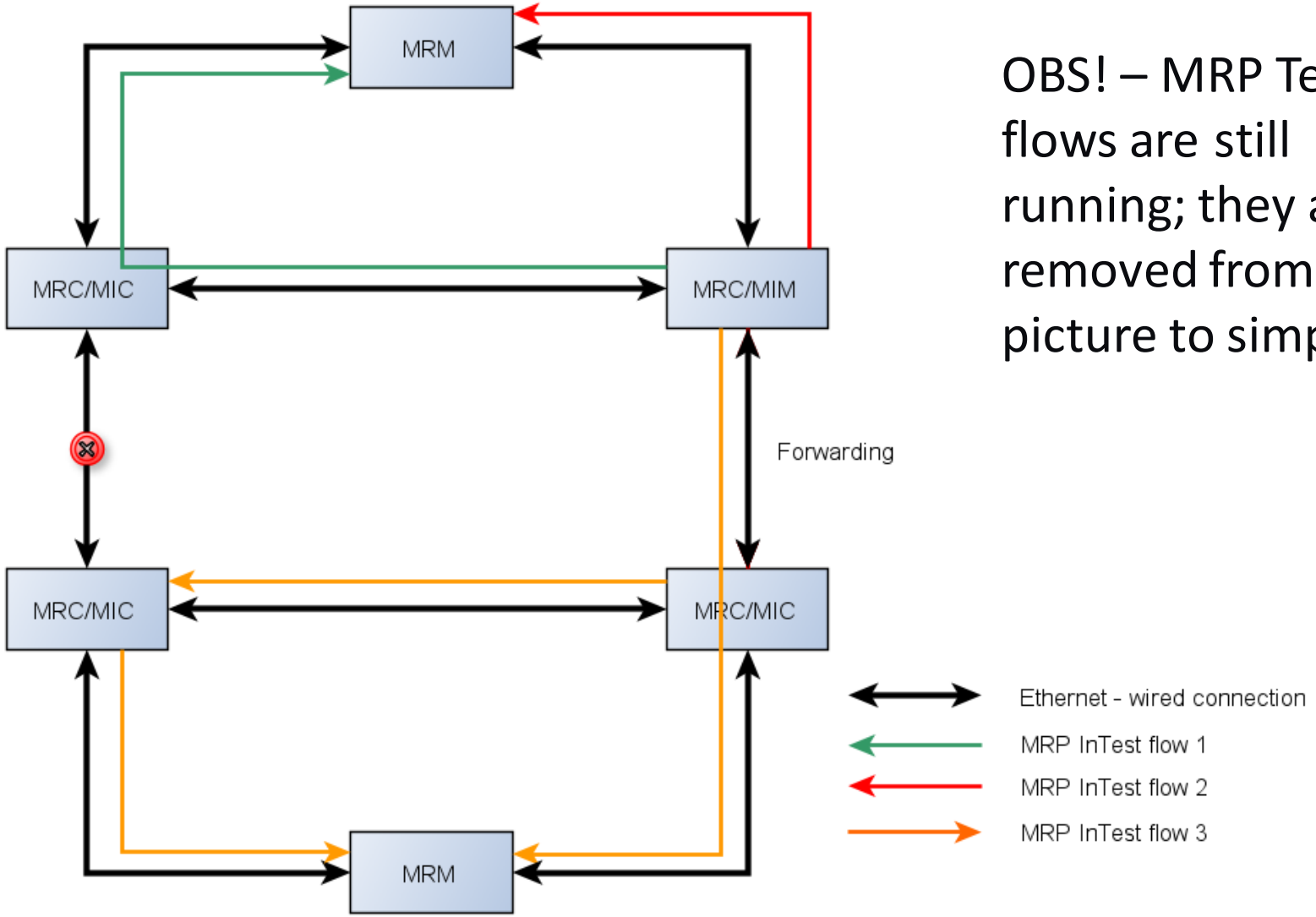
MIM/MIC – RC



OBS! – MRP Test flows are still running; they are just removed from picture to simplify it.

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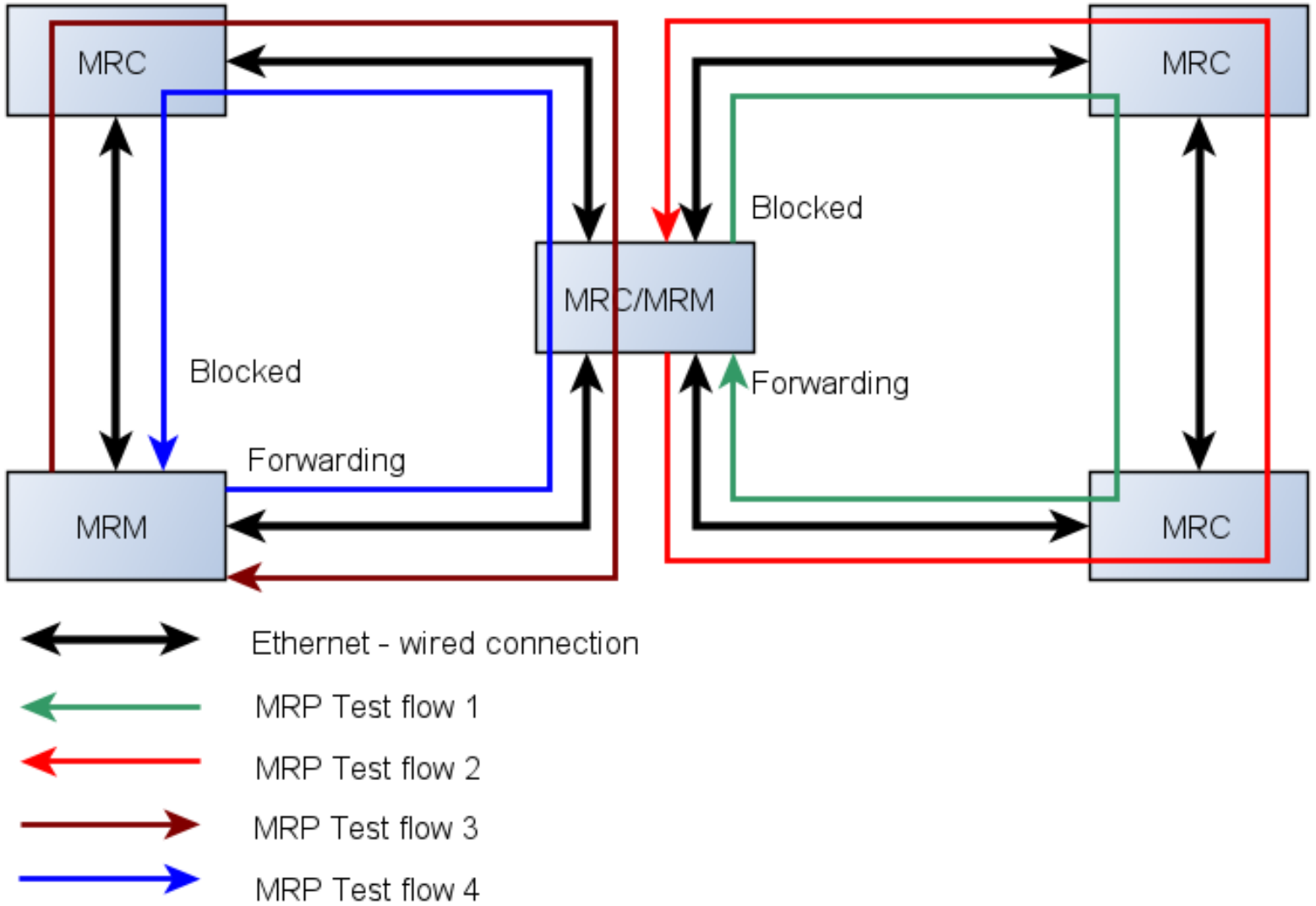
MIM/MIC - RC



OBS! – MRP Test flows are still running; they are just removed from picture to simplify it.

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Multiple Instances



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Implementation

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Implementation

- **Userspace:**
  - implements the MRP state machine + generates non Test/InTest frames
  - https://github.com/microchip-ung/mrp

- **Kernel:**
  - generates Test/InTest frames
  - detects when the ring is open/closed
  - forwards all MRP frames
  - src: *net/bridge/br_mrp.c*

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Use cases

- **Pure SW Bridge**
  - All forwarding is done in SW
- **HW Offloaded Bridge - none MRP aware**
  - Forwarding of Test/InTest is done in HW
  - If they can trap MRP frames, then they can support MRP
  - Some (many?) MRP unaware switches can HW offload MRC
    - Some may only be able to do it with 1 ring, other with N rings
- **HW Offloaded Bridge – MRP Aware**
  - The HW we work with can do the following:
    - Generate MRP Test/InTest frames, up too 4k frames per second
    - Detect when stop receiving MRP Test/InTest frames, need process 4k
    - Forward, redirect, terminate MRP Test/InTest frames

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Profiles

		Profile				
		500ms	**200ms**	**30ms**	**10ms**	Meaning
Parameter	**Topology**	20ms	10ms	0.5ms	0.5ms	Topology Change interval
	Topology cnt	3	3	3	3	Topology Change repeat count
	Test short	30ms	10ms	1ms	0.5ms	Test short interval
	Test default	50ms	20ms	3.5ms	1ms	Test default interval
	Test max	5	3	3	3	Test Max miss count
	LinkDown	20ms	20ms	1ms	1ms	Link Down interval
	LinkUp	20ms	20ms	1ms	1ms	Link Up interval
	LinkCnt	5	5	5	5	Link Change repeat count

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Interconnect profiles

		Profile		
		500ms	**200ms**	**Meaning**
Parameter	**In Topology**	20ms	10ms	In Topology Change interval
	In Topology Cnt	3	3	In Topology Change repeat count
	In Test default	50ms	20ms	In Test default interval
	In Test Max	8	8	In Test Max miss count
	In LinkStatusPoll	20ms	20ms	In LinkStatusPoll interval
	In LinkStatusPoll Cnt	8	8	In LinkStatusPoll count
	In LinkDown	20ms	20ms	In Link Down interval
	In LinkUp	20ms	20ms	In Link Up interval
	In Link Cnt	4	4	In Link Change repeat count

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Future work

- **Support for MIM/MIC**

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Questions?

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